

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.6: Transmission Media (Copper, Fibre-Optic, Wireless)

Total Questions: 4

Mark Schemes begin on Page 4

Question 3 | Source: 9618_w24_qp_13 (Q9.b)

- 9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

- (b)** Copper cable can be used to transmit data in a network.

Complete the table by identifying **and** describing **two other** transmission media that can be used to transfer data in the WAN.

Transmission medium	Description
..... 	
..... 	

[4]

Question 4 | Source: 9618_s23_qp_13 (Q2.d)

- 2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

- (d) Students can connect their devices to the university network using cables or a wireless connection.

Explain the benefits to the students of allowing them to use **both** wired and wireless connections.

[4]

[4]

Question 5 | Source: 9618_w22_qp_13 (Q7.b)

- (b) Give **one** advantage and **two** disadvantages of transmitting data using satellites instead of copper cables.

Advantage 1

.....

Disadvantage 1

.....

Disadvantage 2

.....

[3]

Question 6 | Source: 9618_s21_qp_11 (Q4.c.ii)

- 4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

- (c) Melinda connects her laptop to the internet through her router.

- (ii) Melinda mainly uses the internet to watch films and play computer games.

Tick (✓) **one** box to identify whether Melinda should connect to the router using a wired or wireless network **and** justify your choice.

Wired	
Wireless	

Justification

.....

.....

.....

.....

..... [3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q4.a)

4(a)	1 mark for each benefit, max 3 • Devices can be more mobile as no physical connection is required • No cabling is needed so easier to set up • Easier to add additional devices • Multiple different device types can be connected	3
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q4.b)

4(b)	1 mark for a drawback • Can be less secure • Transmission speed can be slow • Interference is possible • The signal degrades quickly without repeaters/boosters	1
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Mark Scheme for Question 3 | Source: 9618_w24_ms_13 (Q9.b)

9(b)	1 mark for correct media 1 mark for matching description e.g. • Fibre optic • Transmits data as pulses of light • Radio waves / microwaves • Transmits electromagnetic waves on different frequencies	4
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Mark Scheme for Question 4 | Source: 9618_s23_ms_13 (Q2.d)

2(d)	1 mark each to max 4 Examples: • Some students might only have one sort of connection on their device • Wired provides better performance for the student's device • ... for example, enabling faster access to university databases • There will be less interference if students connect via a cable • Students can transmit private/confidential data/work securely • ... for example, their final dissertation • Wireless connection means that the students can use their devices in different rooms/sites/outside/anywhere more freely // student devices can be portable • Wireless connection enables the students to bring multiple devices // bring their own devices // change devices	4
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Mark Scheme for Question 5 | Source: 9618_w22_ms_13 (Q7.b)

7(b)	1 mark for a correct advantage: - not fixed to a single location - allows access in remote / rural areas 1 mark for each correct disadvantage (max 2): - high latency / lag / slow to connect - more expensive than wired methods, as need extra equipment - signal is affected by bad weather - the transmission speed is slower than fixed line broadband - direct line of sight needed	3
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Mark Scheme for Question 6 | Source: 9618_s21_ms_11 (Q4.c.ii)

4(c)(ii)	1 mark per bullet point for justification up to max 3 No mark for identification of wired/wireless Wired - Faster connection // higher bandwidth needed as she is downloading/streaming large files ... less time waiting / less latency / fewer delays More reliable / stable connection ... is less susceptible to issues with distance/walls/interference More secure Wireless - Freedom of movement ... can move between different rooms with a mobile device and still receive/transmit data ... no need of a physical connection - Easily expanded if friends want to access the same network - Less cabling / expertise is needed ... making the initial setup less expensive	3
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